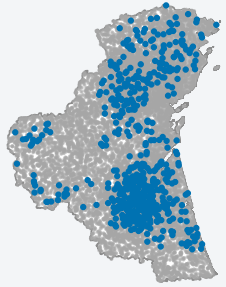
A flexible spatial model for evaluating interventions during avian influenza outbreaks

CONTEXT

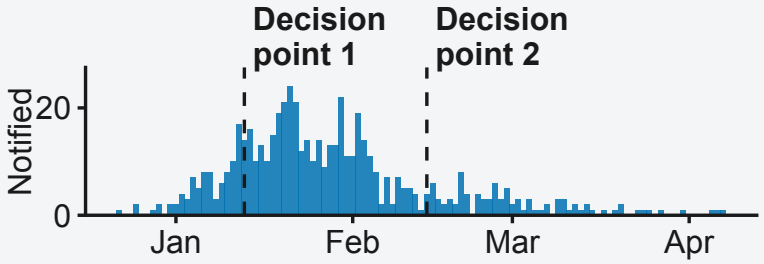


WiLiMan-ID HPAI Modelling Challenge

Synthetic outbreak

9,160 poultry premises

560 notified premises



APPROACH

Spatially explicit stochastic model

S → E → I → N → R

Fit to outbreak data (MCMC)

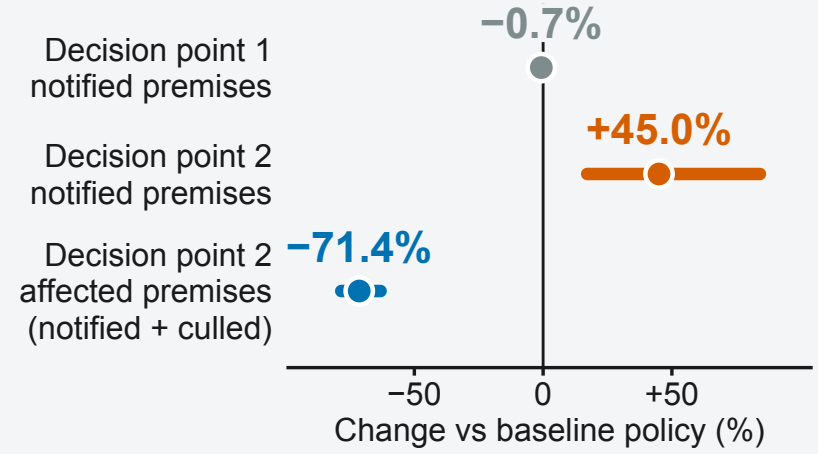
Simulations with interventions for 28 days from each of two decision points

Interventions compared

- Reactive culling
- Preventive culling
- Confinement
- Restocking bans

FINDINGS

With no preventive culling and faster reactive culling:



1. **The effectiveness of interventions can vary between outbreak stages.**
2. **The preferred intervention depends on policy-makers' stated objectives.**